Reg. No. | | | | | | | | | | | [22BSMA11]

FIRST SEMESTER B.Tech EXAMINATION, FEBRUARY - 2023

CALCULUS, LAPLACE TRANSFORMS AND MATRICES

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i.** Missing data may be suitably assumed.
ii. Answer All Questions

PART - A

5 X 2=10

1.	**a.**	Find first order partial derivatives of the function $z = \log(x^2 + y^2)$	L1	CO1	**02**
	b.	Find components of velocity and acceleration at $t = 2$ on the curve $\vec{r} = (t^2+1)\hat{i} + (4t-3)\hat{j} + (2t^2-6t)\hat{k}$	L1	CO1	**02**
	c.	Find Laplace Transform of the function $f(t) = \cos^2 3t$	L1	CO3	**02**
	d.	Find the Inverse Laplace Transform of the function $\frac{1+e^{-3s}}{s^2}$	L1	CO4	**02**
	e.	Define rank of a matrix and hence find the rank of the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$	L1	CO1	**02**
		PART - B			**6 X 5=30**
2.	**a.**	If $u = x^3 - 3xy^2 + x + e^x \cos y + 1$, *then show that* $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$	L2	CO2	**05**
		OR			
	b.	Find total derivative of $z = xy^2 + x^2 y$ *where* $x = at, y = 2at$ by using partial differentiation and hence verify the result by direct substitution.	L2	CO2	**05**
3.	**a.**	If $x = t^2 + 1, y = 4t - 3, z = 2t^2 - 6t$ represents the parametric equation of a curve, then find **i.** unit tangent vector at any point on the curve **ii.** angle between the tangents at $t = 1$ and $t = 2$.	L3	CO6	**05**
		OR			
	b.	Compute $\vec{A}.gradf$, where $f(x,y,z) = 2x + yz - 3y^2$ and $\vec{A} = 3xi + 4yj$.	L3	CO6	**05**
4.	**a.**	Find Laplace transform of the function $t e^{-2t} \sin 4t$	L3	CO3	**05**
		OR			
	b.	Find Laplace transform of the function $\frac{\cos at - \cos bt}{t}$	L3	CO3	**05**
5.	**a.**	Find Inverse Laplace Transform of the function $\log\left[1 - \frac{a^2}{s^2}\right]$	L3	CO4	**05**
		OR			
	b.	Find Inverse Laplace transform of the function $\frac{1}{s(s+1)(s+2)(s+3)}$	L3	CO4	**05**

6.	**a.**	Find rank of the matrix $\begin{bmatrix} 2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$	L3	CO2	**05**
		OR			
	b.	Solve the following system of equations using Gauss-elimination method: $2x + y + z = 10$, $3x + 2y + 3z = 18$ and $x + 4y + 9z = 16$.	L2	CO2	**05**
7.	**a.**	Show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$, where $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$.	L2	CO2	**05**
		OR			
	b.	Find the directional derivative of the function $f(x,y,z) = 4xz^3 - 3x^2y^2z$ at the point (2,- 1, 2) along $2\hat{\imath} - 3\hat{\jmath} + 6\hat{k}$	L3	CO6	**05**
		PART - C		**6 X 10=60**	
8.	**a.**	Examine the function $f(x,y) = x^2y^2 - 5x^2 - 8xy - 5y^2$ for extreme values.	L3	CO6	**10**
		OR			
	b.	**i.** Explain any five elementary MATLAB built-in mathematical functions. **ii.** Write MATLAB code for $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2}$, given $f(x,y) = x^3 + 3xy - 15x^2 - 15y^2 + 72x - 4y$.	L3	CO2	**10**
9.	**a.**	Show that the vector point function $\vec{v} = \frac{x\hat{\imath}+y\hat{\jmath}}{x^2+y^2}$ is both solenoidal and irrotational.	L3	CO6	**10**
		OR			
	b.	**i.** Describe vector built-in functions in MATLAB that operate essentially on vectors returning a scalar value with description. **ii.** Write MATLAB code to draw two-dimensional vector field for the vector $x\hat{\imath} + y\hat{\jmath}$.	L3	CO6	**10**
10.	**a.**	If $f(t) = \begin{cases} \cos t, & 0 < t < \pi \\ \cos 2t, & \pi < t > 2\pi \\ \cos 3t, & t > 2\pi \end{cases}$ express $f(t)$ in terms of unit step function and hence find its Laplace transform.	L3	CO3	**10**
		OR			
	b.	Given that $\frac{dx}{dt} + 4y = 0, \frac{dy}{dt} - 9x = 0$ *and* $x(0) = 2, y(0) = 1$, use Laplace transform to solve the given system of simultaneous differential equation.	L3	CO4	**10**
11.	**a.**	Solve by using Laplace transforms, $y''' + 2y'' - y' - 2y = 0, \quad y(0) = y'(0) = 0, \quad y''(0) = 6$	L3	CO4	**10**
		OR			

	b.	**i.** Explain Laplace and Inverse Laplace transforms Built-in Functions in MATLAB. **ii.** Write MATLAB code for Evaluation of integrals using Laplace Transforms.	L3	CO4	**10**
12.	**a.**	Investigate the values of λ and μ so that the system of equations $x+y+z=6$, $x+2y+3z=10$ and $x+2y+\lambda z=\mu$ will have **i.** no solution **ii.** unique solution **iii.** infinite number of solutions	L3	CO2	**10**
		OR			
	b.	Write MATLAB code to create the matrix $A=\begin{bmatrix}1 & 0 & 4\\ 2 & 8 & 1\\ 5 & 2 & 3\end{bmatrix}$ and also to perform following operations: **i.** Determinant of A **ii.** Rank of A **iii.** Size of matrix A **iv.** Extract 2nd column	L2	CO5	**10**
13.	**a.**	Find a matrix P which diagonalizes the matrix $A=\begin{bmatrix}-19 & 7\\ -42 & 16\end{bmatrix}$. Verify $P^{-1}AP=D$.	L3	CO5	**10**
		OR			
	b.	Diagonalize the matrix $\begin{bmatrix}7 & -2 & 0\\ -2 & 6 & -2\\ 0 & -2 & 5\end{bmatrix}$.	L3	CO5	**10**